FinSynth AI

Multi-Agent Synthesis for Equity Research

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Motivation

1

Reduce Information Overload

Investors are currently drowning in a vast pool of information that is nearly impossible to keep up with.

2

Remove the Financial Barrier of Investing

Professional level investing tools such as Bloomberg terminals provide news, analysis, and trading capabilities, but at a steep price of 24k+/year

3

Allow for quick insights

By the time a human is able to read an annual report or find all the relevant news to invest in a stock, the market has often already priced in the info

Goal

- Develop financial agent that is able to create in depth financial reports within minutes
- Ensure the agent provides accurate financial data and is not prone to hallucination
- Create easy to use frontend to display results and interact with agent

Related Works

TradingAgents: Multi-Agents LLM Financial Trading Framework

- assigns roles inspired by professional firms, including fundamental, sentiment, news, and technical analysts, bullish/bearish researchers and risk managers
- found that agents who debate each other (bull vs bear debate) produce significantly more accurate risk assessments than a single agent

FinRpt: Dataset, Evaluation System and LLM-based Multi-agent Framework for Equity Research Report Generation

- proposes FinRpt benchmark which includes an dataset of 6,825 reports and an 11 metric evaluation system
- developed FinRpt-Gen, an agentic based model that includes 9 specialized agents over 3 modules: Information extraction module, Analysis module, Prediction module
- FinRpt-Gen framework significantly outperformed standalone LLMs, including top tier models (when the paper was written) like GPT-4o and Gemini-2.5-Pro

Implementation

Backend

- MCP Server: yfinance , brave search news or yfinance news
- Agents: Auditor, News Hound, Synthesis
- Uses langraph for orchestrating agent process
- two main APIs consisting of a health check and an analysis run

Frontend

- The frontend was created to use Bloomberg Terminal as a base and uses many of the same colors
- Components: progress indicator, ticker search bar, output box, interactive chat
- Tools: Nextjs, Shadcn, Tailwind CSS

Data Summary

- There is no dataset that is being directly inputted into FinSynth
- Since we do pull data from yfinance via MCP, below is a snapshot of what that data looks like

```
Company     : Coinbase Global, Inc. (COIN)
Sector      : Financial Services / Financial Data & Stock Exchanges
Price       : $181.48
Market Cap  : $48,937,459,712
P/E (trail) : 40.782024
P/E (fwd)   : 30.408846
52-wk range : $139.36 – $444.65
```

Year	Revenue	Rev Gr%	Gross%	Op%	Net%	D/E	CR
2025	$7.2B	+9.4%	74.6%	20.3%	17.6%	1.01	2.34
2024	$6.6B	+111.2%	74.8%	34.1%	39.3%	1.19	2.28
2023	$3.1B	-2.7%	63.5%	-1.7%	3.0%	1.35	2.07
2022	$3.2B	N/A	80.3%	-61.0%	-82.2%	15.45	1.07

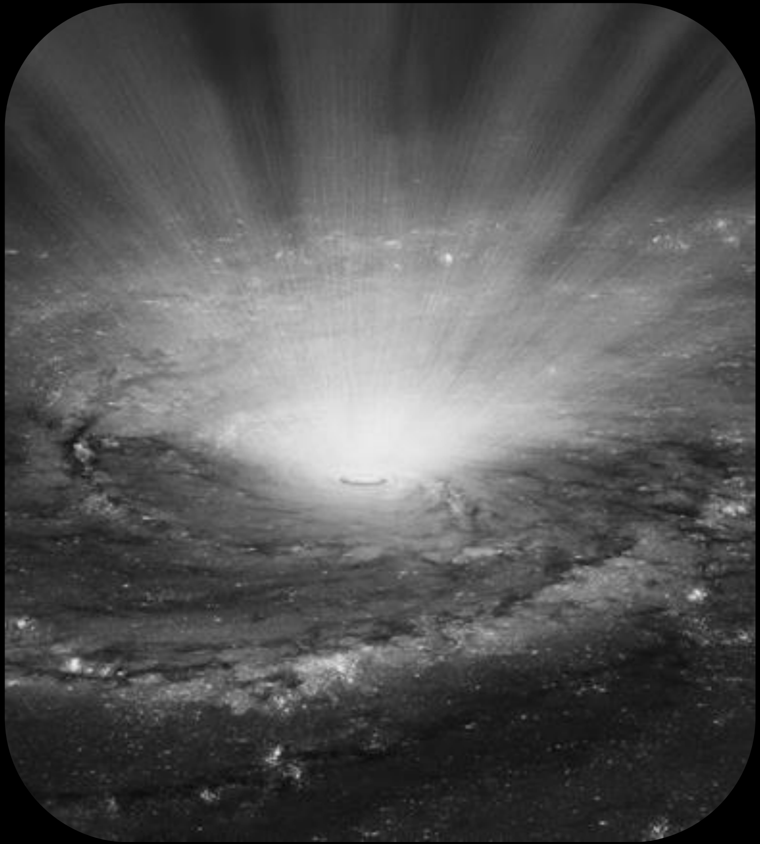
Full JSON output:


```
{
  "ticker": "COIN",
  "company_name": "Coinbase Global, Inc.",
  "sector": "Financial Services",
  "industry": "Financial Data & Stock Exchanges",
  "current_price": 181.48,
  "market_cap": 48937459712,
  "pe_ratio": 40.782024,
  "forward_pe": 30.408846,
  "dividend_yield": null,
  "fifty_two_week_high": 444.65,
  "fifty_two_week_low": 139.36,
  "metrics": [
    {
      "year": "2025",
      "revenue": 7181325000.0,
```

Verifying Financial Claims in Agentic Systems

Approach

- Test direct input of of financial json data to the synthesizer node
- Test the addition of an optional Resynthesizer agent after running a fact check
  - Fact check will draw numbers from the final synthesis report and check to see they match the original MCP data
- Add a UI panel that displays the safety and accuracy of the LLM's analysis.



Mode comparison evaluation

Dataset

- 15 stock tickers split by data richness
- 5 S&P 500 large-caps with rich data and many verifiable numbers
- 5 mid-cap US stocks with moderate data availability
- 5 small-cap stocks with low data

Testing

- Run all three modes on every ticker for 45 total runs
- Measure citation density, total latency, and fact accuracy

Success Metrics

- Achieve greater fact accuracy on both the brief and extra modes

Experimental Results

1. PER-MODE SUMMARY (all tickers combined)

Mode	N	Total(s)	Density	Verified	Unverified	Words	Resynth
brief	15	119.6 ± 11.2	66.0% ± 16.3%	19.1 ± 6.0	10.3 ± 6.4	613 ± 96	0/15
normal	15	203.0 ± 12.2	84.0% ± 12.1%	12.9 ± 2.8	2.4 ± 1.7	834 ± 88	0/15
extra	15	204.0 ± 10.1	84.2% ± 11.1%	13.3 ± 2.8	2.5 ± 1.7	823 ± 98	0/15
extra_force	15	255.7 ± 16.7	84.6% ± 12.0%	12.6 ± 3.3	2.2 ± 1.8	662 ± 99	15/15

2. CITATION DENSITY BY MODE × DATA-RICHNESS TIER (mean ± std)

Tier	Brief	Normal	Extra	ExtraForce
Large-cap	75.6% ± 6.9%	94.6% ± 5.6%	95.8% ± 3.9%	98.3% ± 3.7%
Mid-cap	57.3% ± 24.4%	76.9% ± 11.5%	79.6% ± 9.6%	78.8% ± 9.4%
Small-cap	64.9% ± 8.8%	80.4% ± 11.3%	77.3% ± 8.2%	76.7% ± 6.9%

Experimental Results

3. TOTAL LATENCY BY MODE × TIER (seconds, mean ± std)

Tier	Brief	Normal	Extra	ExtraForce
Large-cap	120.9 ± 11.6	206.3 ± 15.3	204.3 ± 11.8	260.9 ± 21.3
Mid-cap	126.1 ± 12.8	197.0 ± 5.6	206.7 ± 13.5	258.1 ± 16.8
Small-cap	111.9 ± 3.9	205.7 ± 13.8	200.9 ± 3.2	248.2 ± 11.4

6. DENSITY IMPROVEMENT FROM RE-SYNTHESIS (triggered runs only)

```
  Triggered runs:         15
  Draft density:          84.6% ± 12.0%
  Post-resynth density:   80.0% ± 41.4%
  Mean improvement:       -4.6% ± 39.8%

  By tier:
    Large-cap   : 1.7% ± 3.7% over 5 triggered run(s)
    Mid-cap     : 21.2% ± 9.4% over 5 triggered run(s)
    Small-cap   : -36.7% ± 57.1% over 5 triggered run(s)
```

Experimental Analysis

- Extra and normal are functionally identical
- The auditor LLM step provides a notable accuracy improvement
  - Brief generates more numerical claims but less accurately
- Brief's synthesizer took longer than normal's
- Resynthesis is overcorrecting and causing the model to not include enough figures
- The baseline was surprisingly the best performing overall

Conclusion and Future Work

- shows that a small well structured multi agent system can automate a task that requires significant analyst time at a simple level
- By using a three agent system combined with a MCP connection, FinSynth is able to gather accurate financial data and news to create a comprehensive report on the current financial state of a company
    - The original 3 part system was most effective

Future Work

- Integrate MCP connections with more power such as Financial Modeling Prep or Polygon io to get more reliable institution grade financial information
- Add more agents that provide specific viewpoints and integrate them into synthesis
- Change from a linear sequence to parallel to make system faster
- Add self critique loops/ an evaluator agent
- Modify the resynthesis prompt to better retain facts

Links

Github PR: https://github.com/Qdata4Capstone/uva-26-spring-gen-agent-students-projectA/pull/11

Youtube Link: https://youtu.be/AGJMEqU6QNo

Thank you